

Mark Smalley

CTO · Head of Engineering · Blockchain Protocol Architect

Available for the right opportunity · Open to employed roles, UK/Europe & remote globally

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PROFILE

Engineering leader with 15 years building and shipping blockchain infrastructure across Bitcoin, Ethereum, and emerging protocol layers. Proven track record of taking teams from zero to production at high-growth startups — including personally prototyping Ordzaar, which became CakeDefi's fifth business vertical, and leading the engineering pivot at Oviato that secured investment from DraperVC. Currently running an AI-augmented games studio that shipped four products in under six months using LLM-driven engineering workflows. Equally comfortable writing cryptographic code and managing cross-functional teams. Deep expertise in UTXO scripting, Ordinals, passkey/WebAuthN infrastructure, protocol design, and AI-assisted product delivery. Seeking a pre-seed to Series B engineering leadership role where technical depth and startup operator experience both matter.

CORE TECHNICAL SKILLS

Bitcoin / UTXO	UTXO scripting, Schnorr signatures, Taproot, Ordinals, Inscriptions, Runes, BRC-20, BitcoinJS
Ethereum / EVM	Solidity, EthersJS, Web3.js, ERC-20, ERC-721, IPFS, smart contract architecture
Auth & Identity	Passkeys, WebAuthN, FIDO2, cryptographic key management, DN-Key protocol
Backend	Node.js, MongoDB, Neo4j, REST & SDK design, protocol specification, open-source libraries
Frontend	React, Svelte, VueJS, Tailwind, ThreeJS, WebGL — full-stack delivery
AI & Tooling	LLM-augmented engineering workflows, AI game opponents, procedural generation, prompt engineering
Leadership	Hiring, team structure (P4→M3), roadmap ownership, moonshot team formation, VC due diligence

WORK EXPERIENCE

Chief Blockchain Officer Oviato — United Kingdom

Feb 2025 - Mar 2026

Co-founded and led the technical pivot from consumer app to passkey-native wallet infrastructure provider — building on prototype work from Project Eden (weekend hackathon → production architecture). The pivot secured DraperVC investment.

- ▶ Designed and built internal cryptographic framework underpinning the passkey wallet architecture
- ▶ Produced passkey/WebAuthN prototypes that formed the basis of the investor pitch to DraperVC
- ▶ Instrumental in articulating the technical pivot from consumer app to B2B infrastructure play

CEO & Founder (active) Moddable Limited — United Kingdom

2025 - present

Open-source games studio using AI-augmented engineering workflows to ship at startup speed. Four games, two open-source engines, and a tools platform — all live and public within six months.

- ▶ Built two open-source game engines (64-variant chess engine + hex map framework) with developer docs, embed APIs, and procedural generation systems
- ▶ Shipped Dungeon Chess with custom AI opponent architecture — heuristic-driven tactical decision engine for asymmetric play
- ▶ Engineered a custom PDF pagination system — element-level height calculation, orphan prevention, minimum page-fill enforcement, and multi-section document assembly producing 190+ page rulebook libraries
- ▶ Adopted LLM-driven development as core methodology — using AI pair programming to achieve solo-founder velocity across 10+ concurrent repositories

Head of Engineering CakeDefi — Singapore

Feb 2023 - Sep 2023

Joined as P4 Engineer; promoted to M3 Manager within months. Tasked with designing novel Bitcoin meta-protocols, then given mandate to form and lead a new Moonshot team whose first output became Cake's fifth business vertical.

- ▶ Sole developer of the first working prototype for Ordzaar — Bitcoin's leading Ordinals marketplace
- ▶ Founded and structured the Moonshot team: hiring, sprint cadence, product vision, and stakeholder reporting
- ▶ Deepened UTXO expertise across Schnorr signatures, Taproot, Ordinals, Inscriptions, and Runes
- ▶ Co-authored the SADO Protocol — a decentralised, self-authenticating order-book standard for Bitcoin Ordinals — published as an open-source SDK now used across the ecosystem

CEO & Founder R1 — Malaysia

2012 - 2025

Founded and operated Malaysia's leading blockchain infrastructure and advisory firm for 13 years. Built and shipped production products across KYC, AML, digital custody, and wallet infrastructure while simultaneously shaping national regulatory policy.

- ▶ First company admitted to the Securities Commission Malaysia regulatory sandbox
- ▶ Contributed directly to Malaysia's Digital Asset regulatory roadmap alongside SC and central bank stakeholders
- ▶ Co-founded and funded CoKeeps — Malaysia's first regulated digital asset custodian — providing five years of capital from R1
- ▶ Designed and shipped multiple production systems: KYC/AML pipelines, blockchain wallets, learning management, and digital identity platforms — full-stack from protocol to UI
- ▶ Authored DN-Key protocol for decentralised key management and SADO for ordinals order-book infrastructure
- ▶ Enterprise clients included DBS Bank (Singapore) and Baker Hostetler (USA)
- ▶ Shareholder in CoKeeps and Sinegy following equity arrangements from advisory and build work

Chief Technology Officer Neuroware Inc — USA / Malaysia

2013 - 2018

500 Startups Batch 9 (US\$300K+ raised, launched at TechCrunch Disrupt London). Blockchain infrastructure company providing compliance-grade tooling to licensed FinTech operators and regulators across Southeast Asia.

- ▶ Ranked Top 5 by VentureBeat from 500 Startups Batch 9; the only Malaysian company in Deloitte's Top 50 Global RegTech
- ▶ Built early-stage open-source blockchain wallet and SDK infrastructure — handling 1M+ API requests during DBS hackathon deployments
- ▶ Developed 30+ Ethereum smart contracts for BloqVerse (~20 upgradeable, ~10 immutable) — fully serverless architecture with IPFS-served frontend and no traditional database
- ▶ Core technical stack: UTXO scripting, Solidity, EthersJS, BitcoinJS, IPFS, MongoDB, Node.js

OPEN SOURCE & PROTOCOL CONTRIBUTIONS

- ▶ **SADO Protocol** — co-authored self-authenticating, decentralised ordinals order-book; SDK at github.com/sadoprotocol, used in production by Ordzaar and third-party exchanges
- ▶ **DN-Key Protocol** — decentralised key management standard designed for regulated digital asset environments
- ▶ **Ordit SDK** — indexer, parser, and API toolkit for Ordinals and Inscriptions (github.com/sadoprotocol/ordit)
- ▶ **BackPress** — core contributor to Automattic's BackPress framework (WordPress ecosystem)
- ▶ 20 public repositories at github.com/msmalley covering wallet infrastructure, protocol tooling, and blockchain SDKs

SELECTED ACHIEVEMENTS

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| ▶ DraperVC investment secured (Oviato, 2025) — technical work directly influenced funding decision | ▶ Accepted into 500 Startups Batch 9 and Antler Malaysia Batch 2 |
| ▶ Ordzaar prototype — sole developer; became CakeDefi's fifth business vertical | ▶ Deloitte Top 50 Global RegTech Companies (Neuroware, 2017) |
| ▶ TEDx Sarawak speaker (2017) — 'The Cost of Cash' | ▶ Digerati50 — Malaysia's top 50 digital leaders |

EDUCATION & CREDENTIALS

Accepted — 500 Startups Batch 9	Accepted — Antler Malaysia Batch 2
Registered HRDF Trainer (Malaysia)	Silver Duke of Edinburgh Award
A-Level English — Open Uni, 1998–99	GCSEs / O-Levels — 6 Bs & 1 A, 1990–95

REFERENCES

Shafiu Hussain

CEO, Oviato · shaf@oviato.com

U-Zyn Chua

Founder & CTO, Cake Group (Singapore) · chua@uzyn.com